

Dayeon Kang

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RESEARCH INTERESTS

I am interested in the intersection of computer networking, cloud systems, and security and privacy, with a particular focus on autonomous AI agents. This is related to the following areas: 1) Analyzing security risks that emerges as cloud systems scale, and how AI agents can be fingerprinted and attributed through network-layer signals. 2) Designing privacy-preserving AI agents that leverage cloud and network infrastructure.

Keywords: Security and Privacy on Cloud Systems, Network and Distributed Systems, Serverless Cloud, Autonomous AI Agents

EDUCATIONS

University of Massachusetts Amherst (UMass Amherst), MA, USA Sep. 2023 – Present
MS/PhD Student in Computer Science

- Cumulative GPA: 3.9/4.0, as of Spring 2026
- Advisors: Prof. Amir Houmansadr, Prof. Pubali Datta

Ewha Womans University (Ewha WU), Seoul, South Korea Mar. 2019 – Aug. 2023
B.Eng. in Computer Science and Engineering

- Cumulative GPA: 4.16/4.3, *Summa Cum Laude*

PUBLICATIONS

Dayeon Kang, J. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, “Whose Agent Are You? Multi-Layer Fingerprinting and Attribution of Autonomous Web Agents,” under review at IEEE Symposium on Security and Privacy (*IEEE S&P*), 2027. [\[preprint\]](#)

Dayeon Kang, C. John, A. Houmansadr, E. Pauley, and P. Datta, “Characterizing Latent Configuration Security Risks in Serverless Applications,” under review at ACM Internet Measurement Conference (*IMC*), 2026.

Dayeon Kang, E. Yeboah-Duako, S. Thomas, and P. Datta, “When Apps Outlive Vendors: Security Implications of IoT Abandonware,” under review at ACM Conference on Computer and Communications Security (*CCS*), 2026.

Dayeon Kang, J. Sheffey, M. Wu, P. Datta, and A. Houmansadr, “CensorLess: Cost-Efficient Censorship Circumvention Through Serverless Cloud Functions,” Privacy Enhancing Technologies Symposium (*PETS*), 2026. [\[paper\]](#)

J. Sheffey, A. Zohaib, **Dayeon Kang**, Z. Durumeric, A. Houmansadr, and Q. Wu, “I’ll Shake Your Hand: What Happens After DNS Poisoning,” Free and Open Communications on the Internet (*FOCI*), 2025. [\[paper\]](#)

Y. You, J. Yoon, **Dayeon Kang**, J. Kim, and H. Lee, “CollageMap: Tailoring Generative Fingerprint Map via Obstacle-Aware Adaptation for Site-Survey-Free Indoor Localization,” IEEE Pervasive Computing and Communications (*IEEE PerCom*), 2025. [\[paper\]](#)

J. Yoon, Y. You, **Dayeon Kang**, J. Kim, and H. Lee, “CAMPUSRSSI: Real-World Wi-Fi RSSI Measurements for Indoor Localization,” IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (*IEEE PerCom Workshops*), 2025. [\[paper\]](#)

J. Yoon, Y. You, **Dayeon Kang**, J. Kim, and H. Lee, “GAN-Loc: Empowering Indoor Localization for Unknown Areas via Generative Fingerprint Map,” IEEE International Conference on Sensing, Communication, and Networking (*IEEE SECON*), 2024. [\[paper\]](#)

Dayeon Kang, J. Kim, D. Kim, H. Kim, and K. Chae, “Grape Grade Discrimination System using Step-by-step Deep Learning Model and Drone Path Planning Algorithm,” *Korea Computer Congress*, 2022.

Poster

Dayeon Kang, P. Datta, and E. Pauley, “Poster: Measuring the Risk of Exposed Configurations in Serverless Environment,” IEEE Symposium on Security and Privacy (*IEEE S&P*), 2026. [\[poster\]](#)

RESEARCH EXPERIENCE

The Secure, Private Internet Research Group (SPIN), UMass Amherst

PhD Student (Advisor: Prof. Amir Houmansadr, Co-advisor: Prof. Pubali Datta) Sep. 2023 – Present

- Fingerprinted six different AI web agents’ network patterns and behavioral signals according to their systemic characteristics and user-generated prompts
- Measured risks of latent configurations in 16,530 serverless repositories and analyzed how configurations across different cloud providers, providing applications’ structure complexity, inflate attack surfaces
- Measured and conducted meta-analysis on vulnerabilities caused by obsolete libraries, hijackable domains, and broken data flow in abandoned IoT applications
- Built serverless function-based (e.g., AWS Lambda, Microsoft Azure Functions) cost-efficient censorship circumvention proxy with Node.js

Intelligent Networked Systems Lab, Ewha WU

Undergraduate Researcher (Advisor: Prof. HyungJune Lee) Jul. 2021 – Feb. 2023

- Designed initial system to generate obstacle-aware map by reflecting degree of obstacles for each designated in-building structures
- Created Android application with Android Studio to measure Wi-Fi signal strength (e.g., RSSI)
- Implemented indoor localization purpose GAN using TensorFlow and participated in implementing federated learning model, particularly regularizing heterogeneous models (e.g., VGG, ResNet) sequentially trained on tasks

TEACHING & MENTORSHIP EXPERIENCE

Teaching Assistant (TA), UMass Amherst

- Introduction to Computer and Network Security (COMPSCI 360) Fall 2025
- System Defense and Test (COMPSCI 561) Spring 2024, 2025, 2026
- Introduction to Computer and Network Security (COMPSCI 560) Fall 2024

Peer Instructor (Tutor) in Major Courses, Ewha WU

- Computer Programming & Lab (36339-05) Spring 2022
- Computer Architecture (20493-01) Fall 2021, 2022

Mentor of Undergraduate Research Volunteers (URV), UMass Amherst Summer 2024, Winter 2025

- Instructed research projects undertaken by undergraduates and guided poster presentation

Mentor of 2022 Samsung Junior SW Creation Contest, Samsung SDS Jul. 2022 – Aug. 2022

- Served as mentor to middle school student’s environment-themed software project

INVITED TALKS & SERVICES

- Guest Talk** | Women in Cybersecurity (WiCyS) at Umass Mar. 2025
- Journey to Security Researcher
- Guest Lecture** | UMass COMPSCI 560: Introduction to Computer and Network Security Dec. 2024
- Break the anonymity of Tor network: What is traffic analysis?
- PETS 2027 Tadpole Reviewer**
- External Reviewer of USENIX Security 2026**

HONORS & AWARDS

Honors

- Appenzeller Memorial Scholarship**, International Foundation for Ewha WU, Inc. Mar. 2026
- Jung-Hun Scholarship**, Jung-Hun Scholarship Foundation Mar. 2021 – Aug. 2023
- Honors Scholarship (for academic excellence)**, Ewha WU Spring 2020, Fall 2021, Spring 2022
- Alumni Scholarship**, Ewha WU Spring 2020
- Dean's List**, seven semesters, Ewha WU 2019 – 2022

Awards

- PETS 2026 Stipends**, PETS 2026 May. 2026
- Fourth Place in Capstone Design Project Poster Session**, Ewha WU Jun. 2022
- Third Place in Engineering Capstone Design Contest**, Ewha WU Dec. 2021
- Third Place in Smart Farm & AI Challenge**, Korea Ministry of Agriculture, Food and Rural Affairs Nov. 2021
- Second Place in Software Entrepreneur Contest**, Ewha WU Nov. 2021

TECHNICAL SKILLS

- Proficient** C, Python (with PyTorch, TensorFlow), JavaScript (JS), Docker, MATLAB, $\text{\LaTeX}$
- Intermediate** MySQL, JAVA, C#
- Basic** Assembly Language (MIPS), Android Studio